DD_FA-02 - Network Field Audit

Version: v1.0 **Bundle:** 10_other **Status:** Draft for review **Primary owner:** Network operations / field quality team **Related DDs:** DD_RLM-CFG-01, DD_ILM-CFG-02, DD_WO-01, DD_PROV-INT-01, DD_FA-01, DD_OSR-INSTANCE-01, DD_EM-02, DD_EM-CFG-03, DD_EM-CFG-04, DD_TCK-01, DD_REP-01, FOUNDATION_DROOLS

MVP simplification baseline: FA-01, FA-02, and FA-03 must be implemented as one field-audit capability in the first release, with `audit_type` driving type-specific rules, checklists, and routing. Do not create three services or three independent databases.

1. Purpose

FA-02 defines field audit of outside-plant and access-network assets: HomePass availability, pole/tap/splitter/OLT-port condition, cable path, signal quality, GPS position, photos, and serviceability evidence.

Use FA-02 when:

- network operations needs to verify HomePass readiness before selling;
- installation failures suggest wrong network topology or missing plant;
- a field team must confirm splitter/tap/port condition;
- a mass audit is needed for an area, franchise, or technology region;
- support/NOC needs evidence before changing HomePass sellability;
- reporting needs audit coverage and plant quality metrics.

FA-02 is not the authoritative network inventory. RLM-CFG-01 remains the owner of HomePass/serviceability/topology records. FA-02 records audit evidence and routes approved corrections to RLM or the proper network owner.

2. Scope And Boundaries

Capability	Owner
Network audit campaign/task/evidence/discrepancy	FA-02
HomePass, serviceability, topology, address/port availability	RLM-CFG-01
Technology region/team scope	ILM-CFG-02
Field dispatch and technician execution	WO-01
Provisioning/network read checks	PROV-INT-01
Serialized active customer equipment	OSR-INSTANCE-01 / FA-01
Field users, teams, contractors	EM-02
Permissions	EM-CFG-03
Approval for sellability/topology correction	EM-CFG-04
Customer/support case if customer impacted	TCK-01
Analytics and audit dashboards	REP-01
Configurable audit severity and routing	Drools

Boundary rules:

- FA-02 does not directly update HomePass sellability, port availability, or topology tables.
- FA-02 stores evidence, discrepancies, and requested corrections.
- RLM-CFG-01 owns approved HomePass/topology mutation.
- WO-01 owns technician assignment and field finalization when dispatch is required.
- FA-02 does not replace PROV-INT reconciliation. It may call read/check APIs for evidence.
- EM-CFG-04 is required for risky topology/sellability correction where configured. Camunda is used only when the approval policy mode is `BPMN`.

3. Process Overview

```
flowchart LR
  A["Backoffice creates network audit"] --> B["FA-02 loads HomePass/topology scope"]
  B --> C["Create WO or assign field task"]
  C --> D["Technician captures GPS/photos/measurements"]
  D --> E["FA-02 compares expected vs observed"]
  E --> F{"Discrepancy?"}
  F -- "No" --> G["Close task and mark verified"]
  F -- "Yes" --> H["Open discrepancy"]
  H --> I["Drools severity and routing"]
  I --> J["RLM correction / NOC review / ticket / approval"]
```

4. Status Models

4.1 Audit Status

Status	Meaning
`DRAFT`	Audit campaign/task being prepared.
`SCHEDULED`	Audit is scheduled.
`IN_PROGRESS`	Field capture started.
`RECONCILING`	Observations are being compared.
`PENDING_APPROVAL`	Correction needs approval.
`CLOSED`	Audit completed.
`CANCELLED`	Audit cancelled.

4.2 Discrepancy Status

Status	Meaning
`OPEN`	Difference detected.
`ROUTED`	Route/action selected.
`PENDING_APPROVAL`	Approval required.
`ACTION_CREATED`	RLM/TCK/NOC action created.
`RESOLVED`	Corrective action completed.
`CLOSED`	Discrepancy closed.

5. Data Model

5.1 `network_field_audit_campaign`

What it is for: stores one planned network audit scope.

When to use it: create a campaign for area verification, HomePass readiness audit, post-build quality check, or investigation batch.

Sample meaning: `nfac-001` is a GPON readiness audit for Nairobi West covering 300 HomePass records.

Column	Type	Purpose	Sample
`campaign_id`	TEXT PK	Campaign id.	`nfac-001`
`operator_code`	TEXT	Tenant/operator.	`WIK`
`campaign_type`	TEXT	`HOMEPASS_READINESS`, `POST_BUILD_QA`, `AREA_INVESTIGATION`, `SIGNAL_QUALITY`.	`HOMEPASS_READINESS`
`area_code`	TEXT NULL	Area scope.	`NRB-WEST`
`technology_family`	TEXT NULL	`GPON`, `HFC`, `MIXED`.	`GPON`
`status`	TEXT	Campaign status.	`IN_PROGRESS`
`scheduled_start_at`	TIMESTAMP NULL	Planned start.	`2026-08-15T08:00:00Z`

Column	Type	Purpose	Sample
scheduled_end_at	TIMESTAMP NULL	Planned end.	2026-08-25T17:00:00Z
created_by_user_id	TEXT	Creator.	usr-network-audit
created_at	TIMESTAMP	Creation timestamp.	2026-08-01T09:00:00Z
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- (operator_code, status).
- (operator_code, area_code, status).
- (technology_family, status).

5.2 network_field_audit_task

What it is for: stores one field audit assignment for a HomePass, network node, access port, route segment, or address cluster.

When to use it: create one task per physical location or network element to be checked.

Sample meaning: nfat-001 asks technician usr-tech-020 to verify GPON splitter SPL-NRB-009.

Column	Type	Purpose	Sample
audit_task_id	TEXT PK	Task id.	nfat-001
campaign_id	TEXT NULL	Parent campaign.	nfac-001
operator_code	TEXT	Tenant/operator.	WIK
task_type	TEXT	HOMEPASS, SPLITTER, TAP, OLT_PORT, ROUTE_SEGMENT, SIGNAL_TEST.	SPLITTER
network_ref_type	TEXT	Network object type.	SPLITTER
network_ref_id	TEXT	RLM/network object id.	SPL-NRB-009
homepass_id	TEXT NULL	HomePass if task is HomePass-specific.	HP-GPON-0440
wo_id	TEXT NULL	WO if dispatch-backed.	wo-nfa-001
assigned_to_user_id	TEXT NULL	Field user.	usr-tech-020
assigned_team_id	TEXT NULL	Team/contractor.	team-nrb-network
status	TEXT	Task status.	ASSIGNED
due_at	TIMESTAMP NULL	Due timestamp.	2026-08-16T17:00:00Z
created_at	TIMESTAMP	Creation timestamp.	2026-08-15T08:00:00Z
submitted_at	TIMESTAMP NULL	Submission timestamp.	null
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- (operator_code, status).
- (network_ref_type, network_ref_id).
- (assigned_to_user_id, status).
- (wo_id).

5.3 network_field_audit_expected_state

What it is for: stores the expected RLM/network state at the start of the task.

When to use it: create one row per expected HomePass/node/port/capacity/signal condition so audit results can be compared against a frozen snapshot.

Sample meaning: nfae-001 says splitter SPL-NRB-009 should exist at GPS point -1.2921,36.8219 and have four free ports.

Column	Type	Purpose	Sample
`expected_state_id`	TEXT PK	Expected state id.	`nfae-001`
`audit_task_id`	TEXT	Parent task.	`nfat-001`
`expected_code`	TEXT	Expected item/check.	`SPLITTER_LOCATION`
`expected_value_json`	JSONB	Expected value snapshot.	`{"lat":-1.2921,"lng":36.8219,"freePorts":4}`
`source_system`	TEXT	Owner of source data.	`RLM-CFG-01`
`source_ref_id`	TEXT NULL	Source object id.	`SPL-NRB-009`
`created_at`	TIMESTAMP	Snapshot timestamp.	`2026-08-15T08:02:00Z`

Recommended indexes:

- `(audit_task_id)`.
- `(source_ref_id)`.

5.4 `network_field_audit_observation`

What it is for: stores what the field user found.

When to use it: create rows for GPS, photo evidence, port count, signal measurement, node condition, access route blockage, or any field measurement.

Sample meaning: `nfao-001` records that splitter `SPL-NRB-009` was found, photographed, and measured with acceptable optical power.

Column	Type	Purpose	Sample
`observation_id`	TEXT PK	Observation id.	`nfao-001`
`operator_code`	TEXT	Tenant/operator.	`WIK`
`audit_task_id`	TEXT	Parent task.	`nfat-001`
`observation_code`	TEXT	Observation type.	`OPTICAL_POWER_TEST`
`observed_value_json`	JSONB	Captured value.	`{"rxPowerDbm":-19.4,"freePorts":4}`
`condition_status`	TEXT	`OK`, `DAMAGED`, `MISSING`, `BLOCKED`, `UNKNOWN`.	`OK`
`photo_file_ids_json`	JSONB	Evidence files.	`["file-101"]`
`gps_latitude`	DECIMAL NULL	Captured latitude.	`-1.2920`
`gps_longitude`	DECIMAL NULL	Captured longitude.	`36.8218`
`notes`	TEXT NULL	Field note.	`Splitter accessible and labeled.`
`captured_by_user_id`	TEXT	Field user.	`usr-tech-020`
`captured_at`	TIMESTAMP	Capture timestamp.	`2026-08-16T10:15:00Z`
`offline_client_ref`	TEXT NULL	Mobile idempotency ref.	`mob-nfa-991`

Recommended indexes:

- `(audit_task_id)`.
- `(observation_code)`.
- unique `(operator_code, offline_client_ref)`.

5.5 `network_field_audit_discrepancy`

What it is for: stores mismatch or issue found by the network audit.

When to use it: create one row for missing network object, wrong GPS, low signal, blocked access, port mismatch, HomePass not serviceable, or undocumented plant.

Sample meaning: `nfad-001` records that RLM says a HomePass is sellable but the field audit found no usable port.

Column	Type	Purpose	Sample
`discrepancy_id`	TEXT PK	Discrepancy id.	`nfad-001`

Column	Type	Purpose	Sample
audit_task_id	TEXT	Parent task.	nfat-001
discrepancy_type	TEXT	MISSING_NODE, WRONG_GPS, LOW_SIGNAL, PORT_MISMATCH, NOT_SERVICEABLE, UNDOCUMENTED_PLANT.	PORT_MISMATCH
severity	TEXT	LOW, MEDIUM, HIGH, CRITICAL.	HIGH
status	TEXT	Discrepancy status.	ROUTED
route_action	TEXT NULL	REQUEST_RLM_CORRECTION, CREATE_TICKET, CREATE_NETWORK_REPAIR_WO, NOC_REVIEW, NO_ACTION.	REQUEST_RLM_CORRECTION
routed_ref_type	TEXT NULL	Target action type.	RLM_CORRECTION
routed_ref_id	TEXT NULL	Target action id.	rlm-corr-001
approval_request_id	TEXT NULL	EM-CFG-04 request.	apr-nfa-001
created_at	TIMESTAMP	Creation timestamp.	2026-08-16T10:30:00Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	null

Recommended indexes:

- (audit_task_id, status).
- (discrepancy_type, status).
- (approval_request_id).

6. APIs

6.1 Create Network Audit Campaign

```
POST /api/network-field-audits/campaigns
Content-Type: application/json

{
  "operatorCode": "WIK",
  "campaignType": "HOMEPASS_READINESS",
  "areaCode": "NRB-WEST",
  "technologyFamily": "GPON",
  "scheduledStartAt": "2026-08-15T08:00:00Z",
  "scheduledEndAt": "2026-08-25T17:00:00Z"
}
```

6.2 Create Network Audit Task

```
POST /api/network-field-audits/tasks
Idempotency-Key: nfat-HP-GPON-0440-202608
Content-Type: application/json

{
  "operatorCode": "WIK",
  "campaignId": "nfac-001",
  "taskType": "HOMEPASS",
  "networkRefType": "HOMEPASS",
  "networkRefId": "HP-GPON-0440",
  "homepassId": "HP-GPON-0440",
  "assignedToUserId": "usr-tech-020",
  "createWorkOrder": true
}
```

6.3 Submit Observation

```
POST /api/network-field-audits/tasks/nfat-001/observations
Idempotency-Key: mob-nfa-991
Content-Type: application/json

{
  "observationCode": "OPTICAL_POWER_TEST",
  "observedValue": {
    "rxPowerDbm": -19.4,
    "freePorts": 4
  },
```

```
  "conditionStatus": "OK",
  "photoFileIds": ["file-101"],
  "gpsLatitude": -1.2920,
  "gpsLongitude": 36.8218,
  "capturedAt": "2026-08-16T10:15:00Z",
  "offlineClientRef": "mob-nfa-991"
}
```

6.4 Submit Task

```
POST /api/network-field-audits/tasks/nfat-001/submit
Content-Type: application/json

{
  "submittedByUserId": "usr-tech-020",
  "submittedAt": "2026-08-16T10:30:00Z"
}
```

6.5 Route Discrepancy

```
POST /api/network-field-audit-discrepancies/nfad-001/route
Content-Type: application/json

{
  "requestedByUserId": "usr-network-audit",
  "routeAction": "REQUEST_RLM_CORRECTION"
}
```

7. Rules And Drools

Use Drools for configurable severity and routing.

Recommended packages:

- `rules.field_audit.network.scope`
- `rules.field_audit.network.severity`
- `rules.field_audit.network.routing`
- `rules.field_audit.network.approval`

```
package rules.field_audit.network.severity

rule "Sellable HomePass but no usable port is high severity"
when
  $f : NetworkFieldAuditFact(discrepancyType == "PORT_MISMATCH",
                             homepassSellable == true,
                             usablePortCount == 0)
  $d : NetworkFieldAuditDecision()
then
  $d.setSeverity("HIGH");
  $d.route("REQUEST_RLM_CORRECTION");
end
```

```
package rules.field_audit.network.approval

rule "Disable sellability requires approval"
when
  $f : NetworkFieldAuditFact(routeAction == "REQUEST_RLM_CORRECTION",
                             proposedSellability == "NOT_SELLABLE")
  $d : NetworkFieldAuditDecision()
then
  $d.requireApproval("NETWORK_FIELD_AUDIT_SELLABILITY_CHANGE");
end
```

8. Events

Event	When emitted	Key fields
`NetworkFieldAuditCampaignCreated`	Campaign created.	`campaignId`, `areaCode`
`NetworkFieldAuditTaskCreated`	Task created.	`auditTaskId`, `networkRefType`, `networkRefId`
`NetworkFieldAuditSubmitted`	Observations submitted.	`auditTaskId`, `observationCount`

Event	When emitted	Key fields
`NetworkFieldAuditDiscrepancyOpened`	Difference found.	`discrepancyId`, `discrepancyType`, `severity`
`NetworkFieldAuditDiscrepancyRouted`	Route action created.	`routeAction`, `routedRefType`, `routedRefId`
`NetworkFieldAuditTaskClosed`	Task closed.	`auditTaskId`, `closedAt`

9. Frontend And Module Consumers

Consumer	What it uses
Backoffice network audit console	Campaign/task/discrepancy APIs
Contractor technical mobile app	Assigned audit tasks, observation submit, file upload
RLM operations	Correction requests and evidence
NOC/support	Signal/plant issue evidence
Reporting portal	Audit coverage and plant quality dashboards

10. Infrastructure Recommendations

- Run FA-02 with the operations/field API deployment.
- Use existing WO and mobile app infrastructure for dispatch and offline capture.
- Store photos in foundation file storage.
- Use Kafka/outbox for audit events.
- Keep RLM corrections asynchronous and approval-controlled.

Recommended logical handlers:

Handler	Purpose
`fa02-freeze-expected-state`	Reads RLM/PROV and stores expected snapshots.
`fa02-create-wo`	Creates WO-01 audit dispatch if needed.
`fa02-compare-observations`	Finds mismatches.
`fa02-route-discrepancy`	Runs Drools and creates correction/ticket/review action.

11. Test Scenarios

1. Create HomePass audit task and freeze expected RLM state.
2. Create WO with `FIELD_NETWORK_AUDIT` or equivalent operator catalog job type.
3. Mobile observation is idempotent by offline ref.
4. Matching state closes task.
5. Port mismatch creates discrepancy.
6. Low signal creates NOC review or repair WO.
7. Sellability correction calls EM-CFG-04 when configured.
8. No direct RLM mutation occurs without owning module API.